

# Representations of algebraic structure

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SUSTech, March 2026

# Hungry knights problem

There are twelve hungry knights at a round table. Each of them has a plate with certain amount of food. Instead of eating every minute each knight takes one half of his neighbors servings. They start at 10pm. What can you tell about food distribution in the morning?

What will be if king Arthur joins a table?

Let  $F_i$  be the amount of food on the plate of  $i$ -th knight. Then  $x = (x_1, \dots, x_n) \in \mathbb{R}^n$  is food distribution vector. Denote by  $T$  the rotation operator

$$T(x_1, \dots, x_n) = (x_n, x_1, \dots, x_{n-1}),$$

Then the operator  $\Phi$  which we apply to distribution vector  $x$  every minute is

$$\Phi = \frac{T + T^{-1}}{2}.$$

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$T$  is a diagonalizable with the eigenvalues  $\varepsilon^k$ , where  $\varepsilon$  is the  $n$ th primitive root of 1 and  $k = 1, \dots, n$ . Then  $\Phi$  is diagonalizable with the eigenvalues

$$\left\{ \cos \frac{2\pi k}{n} \mid k = 1, \dots, n \right\}.$$

Choose basis of eigenvectors:  $v_k = (\varepsilon^{-k}, \varepsilon^{-2k}, \dots, \varepsilon^{-nk})$   
 $k = 1, \dots, n$ . Then

$$\Phi v_k = \cos \frac{2\pi k}{n} v_k.$$

Write  $x = a_1 v_1 + \dots a_n v_n$ . We have to calculate

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If  $n$  is odd then  $\lim_{m \rightarrow \infty} \Phi^m x = a_n v_n$ ,  $v_n = (1, \dots, 1)$  and each one will get the same amount

$$a_n = \frac{x_1 + x_2 + \dots + x_n}{n}.$$

If  $n$  is even then  $\lim_{m \rightarrow \infty} \Phi^m x = a_{\frac{n}{2}} v_{\frac{n}{2}} + a_n v_n$ ,  $v_{n/2} = (-1, 1, -1, \dots, 1)$  thus food alternates between even and odd knights

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where

$$a_{\frac{n}{2}} = \frac{x_1 - x_2 + x_3 \cdots - x_n}{n} \quad a_n = \frac{x_1 + x_2 + \dots + x_n}{n}.$$

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Representation theory is a branch of mathematics that studies abstract algebraic structures by representing their elements as linear transformations of vector spaces.

Reference: Pavel Etingof and others: "Introduction to representation theory" <http://www-math.mit.edu/~etingof/replect.pdf>

### Definition

- A representation of an associative algebra  $A$  is a vector space  $V$  equipped with a homomorphism  $\rho : A \rightarrow \text{End } V$ .
- Denote by  $GL(V) \subset \text{End } V$  the linear group of  $V$ . Then a representation of a group  $G$  is a vector space  $V$  equipped with a homomorphism  $\rho : G \rightarrow GL(V)$ .
- We make  $\text{End } V = \mathfrak{gl}(V)$  into a Lie algebra with bracket given by the commutator  $[X, Y] = XY - YX$ . Then a representation of a Lie algebra  $\mathfrak{g}$  is a vector space  $V$  equipped with a homomorphism  $\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ .

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A **subrepresentation** of a representation  $V$  is a subspace  $U \subset V$  which is invariant under all operators  $\rho(a)$ ,  $a \in A$ .

Also, if  $V_1$  and  $V_2$  are two representations of  $A$  then the **direct sum**  $V_1 \oplus V_2$  has an obvious structure of a representation of  $A$ .

A nonzero representation  $V$  of  $A$  is said to be **irreducible** if its only subrepresentations are 0 or  $V$  itself, and **indecomposable** if it cannot be written as a direct sum of two nonzero subrepresentations.

Typical problems of representation theory as follows:

- Classify irreducible representations of a given algebra  $A$ .
- Classify indecomposable representations of a given algebra  $A$ .
- Restrict items 1 and 2 to finite dimensional representations.

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# Representations of finite groups

A map  $\phi$  between two representations  $V$  and  $W$  of  $G$  is a vector space map  $\phi : V \rightarrow W$  such that

$$\rho_g \phi(v) = \phi(\rho_g(v)), \quad \forall v \in V, \forall g \in G.$$

It is easy to prove that  $\text{Ker} \phi$  and  $\text{Im} \phi$  are also  $G$ -modules.

## Theorem (Maschke)

*If  $W$  is a subrepresentation of a representation  $V$  of a finite group  $G$ , then there is a complementary invariant subspace  $W'$  of  $V$ , so that*

$$V = W \oplus W'.$$

Any representation of  $G$  is a direct sum of irreducible representations. Such property is called **complete reducibility**. Infinite groups do not have this property.

# Properties:

- The number of irreducible representations of  $G$  is equal to the number of conjugacy classes, moreover

$$|G| = \sum_i \dim^2 V_i.$$

- The dimension of any irreducible representation of  $G$  divides its order.

## Example

- 1 Any irreducible representation of abelian group is one-dimensional
- 2 Symmetric group  $S_3$  has three irreducible representations.

# Weyl algebra $A_1$ has only infinite-dimensional representations

**Weyl algebra:**  $A_1$  is the associative algebra over a field  $\mathbb{C}$  generated by two elements  $x$  and  $\partial$  subject to the **canonical commutation relation**

$$[\partial, x] = \partial x - x\partial = 1.$$

$A_1$  is defined as the algebra of operators acting on  $\mathbb{C}[x]$ , every element is a linear combination of operators of the form  $a(x) \frac{d^n}{dx^n}$ .

Let  $\rho : A_1 \rightarrow \text{End}(V)$  be  $n$ -dimensional representation, then  $Q = \rho(x)$  and  $P = \rho(\partial)$  are  $n \times n$  matrices. Then

$$I_n = PQ - QP = I_n \implies n = \text{tr}(PQ) - \text{tr}(QP) = 0.$$

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# Representations of quivers

A **quiver** is a finite oriented graph  $Q$ . A **representation** of  $Q$  over a field  $k$  is an assignment of a  $k$ -vector space  $V_i$  to every vertex  $i$  of  $Q$  and of a linear operator  $\varphi_\alpha : V_i \rightarrow V_j$  to every arrow  $\alpha : i \rightarrow j$ .

A morphism of representations  $f : V \rightarrow W$  is a collection of linear operators  $f_i : V_i \rightarrow W_i$  which commute with the operators  $\varphi_\alpha$

$$f_j \varphi_\alpha = \varphi_\alpha f_i.$$

Morphisms  $V \rightarrow W$  form a vector space  $\text{Hom}_Q(V, W)$ , the category of finite-dimensional representations of quiver  $Q$  by  $\text{Rep } Q$ .

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### Example (Jordan quiver)

$$Q_1 : \bullet \curvearrowright \Rightarrow (V, x) \quad x \in \text{Hom}(V, V)$$

Classifying representations of  $Q_1$  is equivalent to classifying linear operators up to a change of basis, or matrices up to conjugacy. Over  $\mathbb{C}$  the classification is given by **Jordan canonical form**.

### Example

$$Q_2 : \bullet \longrightarrow \bullet \Rightarrow (V_1, V_2, x) \quad x \in \text{Hom}(V_1, V_2)$$

By a change of basis in  $V_1, V_2$  any such operator can be brought to the form

$$\begin{bmatrix} I_{r \times r} & 0 \\ 0 & 0 \end{bmatrix}$$



One defines direct sum of representations of  $Q$  and subrepresentations of  $Q$ . Then we can define **irreducible** and **indecomposable** representations of  $Q$ .

### Theorem (Krull-Schmidt theorem)

*Any finite-dimensional representation of a quiver  $Q$  can be written as a direct sum of indecomposable representations, and such decomposition is unique up to reordering.*

Any representation of the quiver  $Q_2$  is isomorphic to a direct sum of the following representations

$$I_{12} : \mathbb{C} \xrightarrow{1} \mathbb{C} \quad S(1) : \mathbb{C} \longrightarrow 0 \quad S(2) : 0 \longrightarrow \mathbb{C}$$

In other words isomorphism classes of indecomposable representations of  $Q_2$  are represented by  $I_{12}$ ,  $S(1)$  and  $S(2)$ ; moreover,  $S(1)$  and  $S(2)$  are only simple representations.

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Classification of irreducible or simple representations is relatively easy task for quivers without oriented cycles

### Theorem

*Let  $Q$  be a quiver without oriented cycles. Then representations  $S(i)$ ,  $i \in I$ , form a full list of simple representations of  $Q$ .*

*Proof:* Assume  $V$  is simple, let  $I' = \{i \in I \mid V_i \neq 0\}$ . Since  $Q$  contains no oriented cycles, there must exist  $i \in I'$  such that there are no edges  $i \rightarrow j$ ,  $j \in I'$ . Thus  $V$  contains a subrepresentation  $V'$  with  $V'_i = V_i$ ,  $V'_j = 0$ ,  $i \neq j$ . Since  $V$  is simple, this implies  $V = V'$  and  $\dim V_i = 1$ .

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# Quivers of finite and tame type

We say that

- $Q$  is of **finite type** if it has finitely many indecomposable representations
- $Q$  is of **tame type** if it has in each dimension finitely many one-parameter families indecomposable representations.

$Q_2$  is quiver of finite type, Jordan quiver  $Q_1$  is of tame type.

## Theorem (Gabriel)

*Both the finite and tame properties of  $Q$  do not depend on the orientation of edges. The connected graphs that yield quivers of finite type are the simply laced Dynkin diagrams, while quivers of tame type are the simply laced extended Dynkin diagrams.*

These graphs arise in a multitude of classical problems in mathematics, such as classification of simple Lie algebras, singularities, platonic solids, reflection groups, etc.

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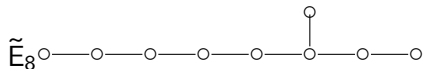
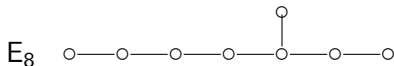
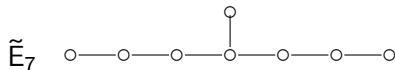
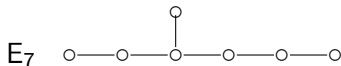
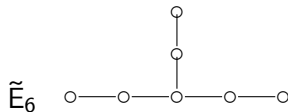
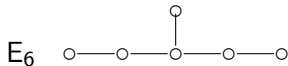
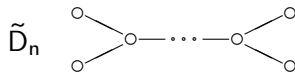
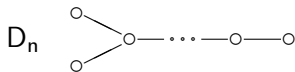
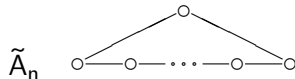
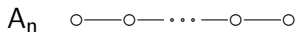
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# (Simply laced) Dynkin and extended Dynkin diagrams



# Path algebra $\mathbb{C}[Q]$

Let  $Q$  be a quiver.

A **path** of length  $r$  in  $Q$  is a sequence of edges such that the source of each next edge coincides with the target of the previous one.

We define **multiplication of paths** by concatenation, then a path of length  $r$  can be written as a product of  $r$  edges.

We introduce paths of length zero  $e_i$ ,  $i \in I$  with  $s(e_i) = t(e_i) = i$ .

## Definition

The path algebra  $\mathbb{C}[Q]$  of quiver  $Q$  is the algebra with the basis given by paths in  $Q$  (including path of length zero) and multiplication given by concatenation.

$\mathbb{C}[Q]$  is finite-dimensional iff the quiver  $Q$  contains no oriented cycles.  $\mathbb{C}[Q]$  is naturally  $\mathbb{Z}_+$ -graded by path length, and  $\mathbb{C}[Q]_0 = \sum_{i \in I} \mathbb{C}e_i$  is semi-simple.

# Path algebra $\mathbb{C}[Q]$

Let  $Q$  be a quiver.

A **path** of length  $r$  in  $Q$  is a sequence of edges such that the source of each next edge coincides with the target of the previous one.

We define **multiplication of paths** by concatenation, then a path of length  $r$  can be written as a product of  $r$  edges.

We introduce paths of length zero  $e_i$ ,  $i \in I$  with  $s(e_i) = t(e_i) = i$ .

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## Theorem

*The category of representations of  $Q$  is equivalent to the category  $\mathbb{C}[Q]\text{-mod}$  of left  $\mathbb{C}[Q]$ -modules.*

*Proof:* Let  $V = (V_i, \varphi_\alpha)$  be a representation of  $Q$ . Define for each path  $p$ , operator  $\varphi_p$  as  $\varphi_{e_i} = \text{Id}_{V_i}$ ,  $\varphi_p = \varphi_{\alpha_r} \dots \varphi_{\alpha_1}$ . Combining we define an action of the path algebra  $\mathbb{C}[Q]$  on the space  $\tilde{V} = \bigoplus_{i \in I} V_i$ .

Conversely, given a  $\mathbb{C}[Q]$ -module  $M$ , define  $M_i = e_i M$ , then for every edge  $\alpha : i \rightarrow j$  in  $Q$  we have  $\alpha(M_i) \subset M_j$ , thus we get a representation of  $Q$ .

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# Representation of Lie algebra $\mathfrak{sl}_2$

A Lie algebra  $\mathfrak{sl}_2$  is a three dimensional algebra, it can be given by three matrices

$$X = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix},$$

and relations  $[H, X] = 2X$ ,  $[H, Y] = -2Y$ ,  $[X, Y] = H$ .

## Definition

Let  $V$  be a representation of  $\mathfrak{sl}_2$ . A vector  $v \in V$  is a **vector of weight**  $\lambda \in \mathbb{C}$  if it is a eigenvector for  $H$  with eigenvalue  $\lambda$   $Hv = \lambda v$ . Denote the subspace of vectors of weight  $\lambda$  by  $V(\lambda)$ .



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### Lemma

If  $v \in V(\lambda)$ ,  $Ev \in V(\lambda + 2)$  and  $Fv \in V(\lambda - 2)$ .

$$\begin{aligned}[H, E] &= HE - EH, & [H, E]v &= HEv - EHv, \\ H(Ev) &= [H, E]v + EHv = 2Ev + \lambda Ev = (\lambda + 2)Ev.\end{aligned}$$

### Theorem

Every finite-dimensional representation  $V$  of  $\mathfrak{sl}_2$  can be written in the form  $V = \bigoplus_{\lambda} V(\lambda)$ .

*Proof:* We prove it when  $V$  is irreducible. Define  $V' = \bigoplus_{\lambda} V(\lambda)$ .  $V'$  is non-zero since  $H$  has at least one eigenvector. Then  $V'$  is non-zero subrepresentation of  $V$ . Then  $V' = V$ .

## Theorem

*Let  $k = \mathbb{C}$ , then the algebra  $\mathfrak{sl}_2$  has exactly one irreducible representation  $V_d$  of each dimension.  $V_d$  is realized in the space of homogeneous polynomials of two variables  $x, y$  of degree  $d - 1$ , and defined by the formulas*

$$\rho(H) = x \frac{\partial}{\partial x} - y \frac{\partial}{\partial y}, \quad \rho(X) = x \frac{\partial}{\partial y}, \quad \rho(Y) = y \frac{\partial}{\partial x}.$$

*Moreover, any indecomposable finite dimensional representation of  $\mathfrak{sl}_2$  is irreducible.*

# Universal enveloping algebra for $\mathfrak{g}$

## Definition

The **universal enveloping algebra**  $U(\mathfrak{g})$  is an associative algebra created by taking the tensor algebra  $T(\mathfrak{g})$  of  $\mathfrak{g}$  quotienting out by the ideal generated by  $x \otimes y - y \otimes x - [x, y]$ ,  $x, y \in \mathfrak{g}$ .

Because of this construction, any action of  $\mathfrak{g}$  on a space  $V$  that satisfies the Lie bracket commutator automatically extends to a valid associative action of  $U(\mathfrak{g})$  on  $V$ :

$$\mathfrak{g} - \text{mod} \simeq U(\mathfrak{g}) - \text{mod}$$

## Example

The set  $\{F^a H^b E^c \mid a, b, c \in \mathbb{Z}^{\geq 0}\}$  forms a basis for  $U(\mathfrak{sl}_2)$ .

# Jordan algebra

Jordan algebras arose from the search of "exceptional" setting for quantum mechanics.

In the usual interpretation - the Copenhagen model -

- the physical observables are self-adjoint or Hermitian matrices (or operators on Hilbert space);
- the basic operations are multiplication by  $\lambda \in \mathbb{C}$ , addition, multiplication of matrices, adjoint operator.

Most of matrix operations are not observable!

In 1932 the German physicist Pascual Jordan proposed a program to discover "a new algebraic setting for quantum mechanics", free from dependence of matrix structure. Jordan has chosen a new observable operation called **quasi-multiplication**

$$x \cdot y := \frac{1}{2}(xy + yx).$$

Next step is to decide empirically what axioms or laws this operation obeys.

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## Definition

A **Jordan algebra** consists of a real vector space equipped with a bilinear product  $x \cdot y$  satisfying the commutative law and the Jordan identity:  $(x^2 \cdot y) \cdot x = x^2 \cdot (y \cdot x)$ ,  $x, y \in J$ . A Jordan algebra is **formally real** if

$$x_1^2 + x_2^2 + \cdots + x_n^2 = 0 \quad \Rightarrow \quad x_1 = x_2 = \cdots = x_n = 0.$$

Any associative algebra  $A$  over  $\mathbb{R}$  gives rise to a Jordan algebra  $A^+$  under quasi-multiplication  $x \cdot y = \frac{1}{2}(xy + yx)$ .

A Jordan algebra is called **special** if it can be realized as a Jordan subalgebra of some  $A^+$ .

If an associative algebra  $A$  with involution  $\star$ , then the subspace of hermitian elements  $H(A, \star) = \{x^\star = x \mid x \in A\}$  forms a special Jordan algebra.



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Another example of quasi-multiplication was found by Jordan on  $J := \mathbb{R}^n \oplus \mathbb{R}1$  for  $n \geq 2$ . Namely 1 acts as identity element and for any  $u, v \in \mathbb{R}^n$  their product is given by inner product in  $\mathbb{R}^n$

$$u \cdot v = \langle u, v \rangle 1.$$

We will call such algebras **Jordan algebras of Clifford type**.

Having settled on the basic axioms for their systems, it remained to find exceptional Jordan algebras Jordan hoped that by studying finite-dimensional algebras he could find families of simple exceptional algebras  $E_n$  parametrized by natural numbers  $n$ , so that letting  $n$  go to infinity would provide a suitable infinite-dimensional exceptional home for quantum mechanics

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In a fundamental 1934 paper Jordan, Wigner and von Neuman showed that every finite-dimensional formally real Jordan algebra is a direct sum of a finite number of the following algebras

- four types of Hermitian  $n \times n$  matrix algebras  $H_n(D)$  corresponding to the four real division algebras  $D$  (the reals, complexes, quaternions, and the octonions or Cayley algebra  $\mathbb{O}$ ) of dimensions 1, 2, 4, 8 (but for the octonions only  $n \leq 3$  is allowed).
- Jordan algebra of Clifford type  $J(\mathbb{R}^n)$ ,  $n \geq 2$ .



Albert showed that  $H_3(\mathbb{O})$  is indeed exceptional Jordan algebras of dimension 27 which now is called **Albert algebra**.

Theorem (Zelmanov, 1983)

*Any simple Jordan algebra, of arbitrary dimension, is either*

- ① *an algebra of Hermitian elements  $H(A, \star)$  for a  $\star$ -simple associative algebra with involution, or*
- ② *an algebra of Clifford type determined by a nondegenerate symmetric form, or*
- ③ *an Albert algebra of dimension 27 over its center.*



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- ③ *an Albert algebra of dimension 27 over its center.*

# Special or one-sided representations of Jordan algebras

We make  $\text{End} V$  into a Jordan algebra  $\text{End} V^+$  with a new product given by  $X \cdot Y = \frac{1}{2}(XY + YX)$ .

## Definition

A **special (or one-sided) representation** of a Jordan algebra  $J$  is a vector space  $V$  equipped with a homomorphism  $\rho : J \rightarrow \text{End} V^+$ .



# Special universal enveloping algebra for $J$

## Definition

The **special universal enveloping algebra**  $U_{\frac{1}{2}}(J)$  is an associative algebra created by taking the tensor algebra  $T(J)$  of  $J$  quotienting out by the ideal generated by  $x \otimes y + y \otimes x - 2x \cdot y$ ,  $x, y \in J$ .

Because of this construction, any action of  $J$  on a space  $V$  that satisfies the anti-commutator relation automatically extends to a valid associative action of  $U_{\frac{1}{2}}(J)$  on  $V$ :

$$J - \text{mod}_{\frac{1}{2}} \simeq U_{\frac{1}{2}}(J) \text{ mod}$$

## Example

$$J = \left\{ \begin{bmatrix} a & b \\ b & c \end{bmatrix} \mid a, b, c \in \mathbb{C} \right\} \implies U_{\frac{1}{2}}(J) = \text{Mat}_{2 \times 2}(\mathbb{C}).$$

Thus  $J - \text{mod}_{\frac{1}{2}}$  contains unique irreducible two-dimensional representation and is semisimple.